Storing accumulators instead of images: exact merging of azimuthally integrated scattering data with uncertainty propagation

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Synopsis

Mathematical development allowing the reduction of scattering data allowing the quality validation, data merging and re-binning with error propagation.

Abstract

Single paragraph stating as specifically and as quantitatively as possible the principal results obtained, and providing an indication of the broader significance of the work. The abstract should be capable of being understood on its own without access to the text or figures.

Keywords: pyFAI, Azimuthal integration, accumulator, spottiness, texture, frame averaging

1 Introduction

2 Mathematical developments

2.1 Azimuthal integration as a weighted mean

In pyFAI, each pixel i provides a signal s_i (raw counts, after dark-current subtraction) and a normalization n_i, the product of the flat-field, solid-angle, polarization and detector-efficiency corrections, possibly multiplied by the incoming flux (Kieffer *et al.*, 2020). A pixel contributes to a bin P with a coefficient $0 < c_i \leq 1$, the fraction of its area falling into the bin, which is 1 without pixel splitting. With the corrected intensity $x_i = s_i/n_i$ and the weight $\omega_i = c_i n_i$, the integrated intensity of the bin is a weighted mean:

$$\widehat{x}_P = \frac{\sum_{i\in P} c_i s_i}{\sum_{i\in P} c_i n_i} = \frac{\sum_{i\in P} \omega_i x_i}{\sum_{i\in P} \omega_i}. \quad (1)$$

Neglecting the uncertainty of the normalization, the variance of this mean is (Kieffer *et al.*, 2020; Kieffer *et al.*, 2025)

$$\sigma^2(\widehat{x}_P) = \frac{\sum_{i\in P} c_i^2\, \sigma^2(s_i)}{\left(\sum_{i\in P} \omega_i\right)^2}. \tag{2}$$

pyFAI offers two models for the variance of the signal. The *Poisson* model propagates a known variance, *e.g.* $\sigma^2(s_i) = s_i$ for a counting detector. The *azimuthal* model assumes that all pixels of a bin sample the same intensity and estimates the variance from their dispersion, $\sigma^2(s_i) \approx n_i^2 (x_i - \widehat{x}_P)^2$, which also works for detectors whose noise is not Poissonian (Kieffer *et al.*, 2025). Both estimates agree on isotropic data recorded with a counting detector (Sedlak *et al.*, 2017). The azimuthal model gives

$$\sigma^2(\widehat{x}_P) = \frac{\sum_{i\in P} \omega_i^2 (x_i - \widehat{x}_P)^2}{\left(\sum_{i\in P} \omega_i\right)^2}. \tag{3}$$

The mean is weighted by ω_i, whereas the squared deviations are weighted by ω_i^2. This is the main difference with the weighted (co-)variance of Schubert & Gertz (2018), where the same weights are used for both.

Two uncertainties are reported. The standard error of the mean, $\mathrm{sem} = \sigma(\widehat{x}_P)$, is the uncertainty on the integrated intensity. The standard deviation, std, is the uncertainty on the intensity of a single pixel. They are related by $\mathrm{sem} = \mathrm{std}/\sqrt{N_{\mathrm{eff}}}$, where $N_{\mathrm{eff}} = \left(\sum \omega_i\right)^2 / \sum \omega_i^2$ is the effective number of pixels in the bin (Kieffer *et al.*, 2025).

2.2 Accumulators

All the quantities above derive from a few sums over the pixels of a bin, called accumulators. Table 1 adapts the notation of Schubert & Gertz (2018) to pyFAI.

Table 1: Accumulators of a bin (or partition) P, adapted from Table 1 of Schubert & Gertz (2018). The last column gives the name of the field in pyFAI integration results.

Symbol	Definition	pyFAI
Ω_P	$\sum_{i\in P} \omega_i$	`sum_normalization`
$\Omega_P^{(2)}$	$\sum_{i\in P} \omega_i^2$	`sum_normalization2`
X_P	$\sum_{i\in P} \omega_i x_i = \sum_{i\in P} c_i s_i$	`sum_signal`
$X_P^{(2)}$	$\sum_{i\in P} \omega_i^2 (x_i - \widehat{x}_P)^2$	`sum_variance`
C_P	$\sum_{i\in P} \omega_i^2 (x_i - \widehat{x}_P)$	new
N_P	$\sum_{i\in P} c_i$	`count`

The intensity and its uncertainties follow from the accumulators:

$$\widehat{x}_P = \frac{X_P}{\Omega_P}, \qquad \mathrm{sem}_P = \frac{\sqrt{X_P^{(2)}}}{\Omega_P}, \qquad \mathrm{std}_P = \sqrt{\frac{X_P^{(2)}}{\Omega_P^{(2)}}}. \tag{4}$$

With the Poisson model, $X_P^{(2)}$ is replaced by the propagated variance $\sum_{i\in P} c_i^2\,\sigma^2(s_i)$, and C_P is not needed (Section 2.3).

The new accumulator C_P is specific to squared weights. With the weights of Schubert & Gertz (2018), its analogue $\sum \omega_i(x_i - \widehat{x}_P)$ vanishes by definition of the mean. Here, $C_P = \Omega_P^{(2)}\,(\tilde{x}_P - \widehat{x}_P)$, where $\tilde{x}_P$ is the mean weighted by ω_i^2. C_P vanishes when all the weights of the bin are equal, but not in general.

2.3 Exact merge in the formalism of Schubert & Gertz (2018)

Let A and B be two disjoint partitions, *e.g.* the same bin in two frames, and $A \sqcup B$ their union. Ω, $\Omega^{(2)}$, X and N are plain sums, hence additive, and the mean is updated as in Schubert & Gertz (2018):

$$\Omega_{A\sqcup B} = \Omega_A + \Omega_B, \quad \Omega^{(2)}_{A\sqcup B} = \Omega^{(2)}_A + \Omega^{(2)}_B, \quad X_{A\sqcup B} = X_A + X_B, \quad \widehat{x}_{A\sqcup B} = \widehat{x}_A + \frac{\Omega_B}{\Omega_{A\sqcup B}}\,(\widehat{x}_B - \widehat{x}_A)\,. \tag{5}$$

With $\Delta x = \widehat{x}_A - \widehat{x}_B$, the sum of squared deviations and the new accumulator are merged exactly by (Appendix A)

$$X^{(2)}_{A\sqcup B} = X^{(2)}_A + X^{(2)}_B + \frac{\Omega^{(2)}_A\,\Omega_B^2 + \Omega^{(2)}_B\,\Omega_A^2}{\Omega^2_{A\sqcup B}}\,\Delta x^2 + 2\,\frac{\Omega_B\,C_A - \Omega_A\,C_B}{\Omega_{A\sqcup B}}\,\Delta x, \tag{6}$$

$$C_{A\sqcup B} = C_A + C_B + \frac{\Omega^{(2)}_A\,\Omega_B - \Omega^{(2)}_B\,\Omega_A}{\Omega_{A\sqcup B}}\,\Delta x. \tag{7}$$

When the same weights are used for the mean and the deviations ($\Omega^{(2)} \to \Omega$, $C \to 0$), equation (6) reduces to equation (21) of Schubert & Gertz (2018). The merge is symmetric in A and B, so it can be applied in any order: sequentially, frame after frame, or as a parallel tree reduction.

The term in C cannot be avoided. Two partitions may share the same $(\Omega, \Omega^{(2)}, X, X^{(2)})$ but have different C, and therefore give different $X^{(2)}$ once merged with the same third partition (Appendix A). No merge formula based on the four accumulators stored by pyFAI so far can thus be exact. The approximate formula used up to now, equation (18) of Kieffer *et al.* (2025), is accurate when the weights are nearly uniform, but its error grows for tree reductions over strongly heterogeneous weights, as shown in the pyFAI tutorial on uncertainties (supporting information).

With the Poisson model, the variance accumulator is a sum of propagated variances, $\sum_{i\in P} c_i^2\,\sigma^2(s_i)$, and is merged by a plain addition, like Ω, $\Omega^{(2)}$ and X. Only the azimuthal model requires C.

Equations (6) and (7) also apply to a partition $B = \{b\}$ made of a single pixel ($X_b^{(2)} = C_b = 0$), which gives the update used when pixels are accumulated one at a time (Appendix A). As for the means, Δx is a difference of close numbers: X and Ω are accumulated in double precision, or in double-word arithmetic on GPU (Joldes *et al.*, 2017), whereas $X^{(2)}$, $\Omega^{(2)}$ and C are well conditioned and single precision is sufficient.

2.4 Scope of the exact merge: pixel splitting

Equations (6) and (7) are exact for the union of the contributions of A and B. They reproduce a direct integration of $A \cup B$ only if no pixel contributes to both partitions. This is always the case for different frames or scan points. It is not the case when neighbouring bins of the same frame are merged, *e.g.* azimuthal sectors into a 1D profile, while pixel splitting is enabled. A pixel split between A and B with the coefficients c_{iA} and c_{iB} then contributes $(c_{iA}^2 + c_{iB}^2)\, n_i^2$ to $\Omega^{(2)}$, $X^{(2)}$ and C instead of $(c_{iA} + c_{iB})^2\, n_i^2$, which underestimates the uncertainty. Pixel splitting also correlates neighbouring bins (Yang *et al.*, 2014). **The merge is therefore rigorously exact only when pixel splitting is disabled**, or when the merged partitions do not share any pixel. With hybrid photon-counting detectors such as Eiger or Pilatus, whose point-spread function is about one pixel without correlation between neighbours, pixel splitting brings little and is best disabled for data meant to be merged, with bins at least as wide as a pixel.

2.5 Renormalization after integration

Since the signal and the normalization are accumulated separately, any correction factor k that is constant over a frame can be applied after integration: the incoming flux measured by a beam-stop diode, the transmission, the exposure time, or the absolute solid-angle scale used when merging different geometries (Kieffer *et al.*, 2020). Multiplying all n_i by k gives

$$\Omega \to k\,\Omega, \qquad \Omega^{(2)} \to k^2\,\Omega^{(2)}, \qquad C \to k\,C, \qquad X,\ X^{(2)} \text{ unchanged}, \tag{8}$$

for both error models. Frames recorded with different monitors are thus renormalized, then merged with their proper weights. Corrections that vary from pixel to pixel within a bin, such as an updated flat-field (Weng *et al.*, 2023), require a new integration of the raw data.

2.6 Derived metrics for data-quality assessment

2.6.1 Spottiness

An ideal powder gives Debye–Scherrer rings of uniform intensity along the azimuth. Large crystallites, a small beam or a short exposure break the rings into spots, which the azimuthal error model measures directly. pyFAI defines a spottiness index as the quadratic mean, over the radial bins r, of the relative standard error of the mean:

$$S = \left(\frac{\sum_r w_r \, (\mathrm{sem}_r / \widehat{x}_r)^2}{\sum_r w_r} \right)^{1/2} = \left(\frac{\sum_r w_r \, X_r^{(2)} / X_r^2}{\sum_r w_r} \right)^{1/2}, \tag{9}$$

with $w_r = 1$, or $w_r = \widehat{x}_r$ to emphasize the Bragg peaks. It is inspired by the normalized r.m.s. intensity variation of Ingham (2014), which is a coefficient of variation, $\mathrm{std}/\widehat{x}$. As $\mathrm{sem} = \mathrm{std}/\sqrt{N_{\mathrm{eff}}}$, S also depends on the number of pixels per bin, hence on the binning, and it decreases as frames are merged: with independent frames, as $1/\sqrt{N}$. S is therefore a measure of the quality of the

integrated pattern, rather than of the sample alone. The coefficient of variation, $\sqrt{X_r^{(2)}/\Omega_r^{(2)}}\,\Omega_r/X_r$, is available from the same accumulators when a measure independent of the binning is preferred. Both only depend on the accumulators and can be calculated after any merge.

2.6.2 Textures

The absence of texture can be checked on the integrated data, using azimuthal sectors. Without texture, the intensities I_j of the M sectors of a radial bin agree within their standard errors, and

$$\chi^2 = \sum_{j=1}^{M} \frac{(I_j - I)^2}{\mathrm{sem}_j^2}, \tag{10}$$

where I is the intensity of the merged sectors, approximately follows a χ^2 distribution with $M-1$ degrees of freedom. Once the absence of texture is established, frame by frame or after merging frames, the sectors are merged with equations (6) and (7) into a 1D pattern. Both the test and this merge assume that pixel splitting is disabled (Section 2.4). Conversely, when texture is the signal of interest, the sectors are kept, and frames with similar orientations are merged sector by sector.

3 Implementation and storage

4 Applications

4.1 Small-angle scattering of macromolecules in solution

BM29 is the BioSAXS beamline of the ESRF, dedicated to small-angle scattering of proteins and other biological macromolecules in solution. Its data are reduced online by the Dahu pipelines (Kieffer *et al.*, 2022), in which the developments described here are already integrated.

4.1.1 Detection of bubbles and menisci

Bubbles in the capillary, or a meniscus when the volume of solution is insufficient, produce streaks on the scattering images. These streaks are easily spotted as a sharp increase of the spottiness (Section 2.6.1). The corresponding frames are automatically flagged as dubious and excluded from further analysis.

4.1.2 Frame averaging

Since the sample is a liquid, its scattering is assumed to be isotropic, and the Poisson error model is used because it provides smoother uncertainty estimates. The accumulators of each frame are saved in the HDF5 file, which allows the signal to be renormalized afterwards, *e.g.* to correct for the noise of the beam-stop diode (Section 2.5). In sample-changer mode, the frames are compared to detect

radiation damage, and only equivalent frames are merged. In HPLC mode, the frames are compared with the first singular vector of the singular value decomposition (SVD) of the chromatogram, and those most similar to it are merged using their accumulators to provide a very precise measurement of the buffer, whereas Kieffer *et al.* (2022) had to average the integrated curves.

4.2 Powder diffraction

Data collected at material science ID11 were - Absence of texture - accumulate how long

4.3 Texture tomography

A dataset of kevlar fibers which was collected at the ID13 beamline at ESRF was integrated and re-binned to a 36x500

5 Discussion and conclusion

A Derivation of the exact merge

Schubert & Gertz (2018) prove their general merge formula through an identity derived backwards. With squared weights, a direct decomposition is simpler. Let $\delta_A = \widehat{x}_A - \widehat{x}_{A\sqcup B}$ be the shift of the mean of A with respect to the mean of the union. Writing $x_i - \widehat{x}_{A\sqcup B} = (x_i - \widehat{x}_A) + \delta_A$,

$$\sum_{i\in A} \omega_i^2 \,(x_i - \widehat{x}_{A\sqcup B})^2 = X_A^{(2)} + 2\,\delta_A\, C_A + \Omega_A^{(2)}\,\delta_A^2, \qquad \sum_{i\in A} \omega_i^2 \,(x_i - \widehat{x}_{A\sqcup B}) = C_A + \Omega_A^{(2)}\,\delta_A. \tag{11}$$

With the weights of Schubert & Gertz (2018), the middle term vanishes since $C_A = 0$. Adding the same expressions for B gives

$$X_{A\sqcup B}^{(2)} = X_A^{(2)} + X_B^{(2)} + \Omega_A^{(2)}\,\delta_A^2 + \Omega_B^{(2)}\,\delta_B^2 + 2\,(\delta_A\, C_A + \delta_B\, C_B)\,, \tag{12}$$

$$C_{A\sqcup B} = C_A + C_B + \Omega_A^{(2)}\,\delta_A + \Omega_B^{(2)}\,\delta_B. \tag{13}$$

From equation (5), $\delta_A = \Omega_B\,\Delta x/\Omega_{A\sqcup B}$ and $\delta_B = -\Omega_A\,\Delta x/\Omega_{A\sqcup B}$, which yields equations (6) and (7). The same decomposition applied to the products $(v_i - \widehat{v})(w_i - \widehat{w})$ gives the merge of weighted covariances, with one accumulator C per variable.

Form using sums. As pyFAI stores sums rather than means, $\Delta x = (\Omega_B X_A - \Omega_A X_B)/(\Omega_A \Omega_B)$ is used in equations (6) and (7), so that the divisions come last (Schubert & Gertz, 2018).

Single pixel. For $B = \{b\}$, with $\Omega_b = \omega_b$, $\Omega_b^{(2)} = \omega_b^2$, $X_b^{(2)} = C_b = 0$ and $X_A - \Omega_A x_b = \Omega_A \, \Delta x$, the update reads

$$X^{(2)}_{A \sqcup \{b\}} = X^{(2)}_A + \omega_b^2 \, \frac{\Omega^{(2)}_A + \Omega^2_A}{\Omega^2_A \, \Omega^2_{A \sqcup \{b\}}} \, (X_A - \Omega_A x_b)^2 + 2 \, \omega_b \, \frac{C_A}{\Omega_A \, \Omega_{A \sqcup \{b\}}} \, (X_A - \Omega_A x_b) \, , \tag{14}$$

$$C_{A \sqcup \{b\}} = C_A + \omega_b \, \frac{\Omega^{(2)}_A - \omega_b \, \Omega_A}{\Omega_A \, \Omega_{A \sqcup \{b\}}} \, (X_A - \Omega_A x_b) \, , \tag{15}$$

the analogue of the update of Youngs & Cramer (1971). The update used so far, equation (17) of Kieffer *et al.* (2025), is the update of Welford (1962) with ω_b^2 in place of ω_b. It misses the factor $(\Omega^{(2)}_A + \Omega^2_A)/(\Omega_A \, \Omega_{A \sqcup \{b\}})$ and the term in C_A, which both vanish when all weights are equal. This explains why it is accurate for nearly uniform weights.

C **cannot be avoided.** Consider the partitions, given as pairs (ω_i, x_i), $A_1 = \{(1,0), (1,2), (2,3)\}$ and $A_2 = \{(1,0), (1,4), (2,2)\}$. Both have $\Omega = 4$, $\Omega^{(2)} = 6$, $X = 8$ and $X^{(2)} = 8$, but $C_{A_1} = 2$ and $C_{A_2} = 0$. Merged with the single pixel $B = \{(1,0)\}$, they give $X^{(2)} = 13.12$ and $X^{(2)} = 11.52$ respectively, as given by equation (6) and by a direct calculation. No function of $(\Omega, \Omega^{(2)}, X, X^{(2)})$ alone can produce both values.

Acknowledgements

The contributions of non-authors etc. should be given here.

Funding

List funding organizations, recipients, grant numbers, etc.

Conflicts of interest: There are no conflicts of interest.

Data availability: Please state how the data supporting the results reported in your article can be accessed, e.g. within the article, as published supporting material, in repositories, upon request...

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